OBJECT ORIENTED PROGRAMMING

ACTIVITY #5 - Personal Expense Tracker

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Instructions:

1) Create a Java program named ExpenseTrackerLastName.
2) Implement the following methods:
   - A void method that displays the program title.
   - A non-void method that calculates the total expenses using three values:
     - Food
     - Transportation
     - other expenses
   - A non-void method that checks if the total expense exceeds a budget and returns a message.
   - A void method that displays the total expense and the budget status.
3) In the main method:
   - Call all the methods you created.
   - Display the final output clearly.
4) Use appropriate method names, parameters, and return types.
5) Add at least TWO enhancements:
     - Accept user input
     - Add another expense category
     - Compute remaining budget
     - Display a personalized message
6) Ensure your program:
     - Compiles without errors
     - Produces correct output

Source Code:

```java
import java.util.Scanner;

public class ExpenseTrackerRivas {

    public static void greetUser() {
        System.out.println(x: "╔═══════════════════════════════════╗");
        System.out.println(x: "╚═══╗ ╔═╗ ╔═╗ ╔═
```

```java
                food = input.nextDouble();

                // Asks for transportation expense
                System.out.print(s: "> How much is your transportation expense?: ");
                transportation = input.nextDouble();

                // Additional Expense Category: Asks for savings expense
                System.out.print(s: "> How much is your savings expense?: ");
                savings = input.nextDouble();

                // Asks for other expenses
                System.out.print(s: "> How much is your other expenses?: ");
                others = input.nextDouble();

                // Go to method to calculate
                sumOfExpense = calculateTotalExpense(food, transportation, savings, others);
                // Prints the sum
                System.out.println("> Here's your total expenses: " + sumOfExpense);
                break;

            case 2:
                System.out.println(x: " === Checking Expense and Budget... === ");
                System.out.print(s: "> Enter current budget: ");
                currentBudget = input.nextDouble();
                budgetStatus = checkExceedBudget(sumOfExpense, currentBudget);

                // Prints budget status
                System.out.println(x: "╔═══════════════════════════════════════╗");
                System.out.println("│ " + budgetStatus + " │");
                System.out.println(x: "╚═══════════════════════════════════════╝");
                break;

            case 3:
                // Prints and display expense and budget
                displayExpenseBudget(sumOfExpense, budgetStatus);
```


```java
                System.out.println(x: " === Checking Expense and Budget... === ");
                System.out.print(s: "> Enter current budget: ");
                currentBudget = input.nextDouble();
                budgetStatus = checkExceedBudget(sumOfExpense, currentBudget);

                // Prints budget status
                System.out.println(x: "╔═══════════════════════════════════════╗");
                System.out.println("│ " + budgetStatus + " │");
                System.out.println(x: "╚═══════════════════════════════════════╝");
                break;

            case 3:
                // Prints and display expense and budget
                displayExpenseBudget(sumOfExpense, budgetStatus);
                break;
```


```java
            case 4:
                // Exits the whole program
                System.out.println(x: " === Exiting System... === ");
                System.exit(status: 0);
                break;
            default:
                // Checks for validation
                System.out.println(x: " WARNING: Please enter valid choice!");
                break;
        }
    }

    }

}
```

Program Output:

```
EXP
EXPENSE TRACKER SYSTEM

> Greetings, User! Welcome to your expense tracker system...
    1. Calculate Total Expense
    2. Check Budget
    3. Display Expense and Budget
    4. Exit

> Please enter your choice:
```

Calculating Total Expense:

```
> Please enter your choice: 1
 === Calculating Total Expense... ===
> How much is your food expense?: 100
> How much is your transportation expense?: 100
> How much is your savings expense?: 300
> How much is your other expenses?: 400
> Here's your total expenses: 900.0
```

Check Expense and Budget:

```
> Please enter your choice: 2
 === Checking Expense and Budget... ===
> Enter current budget: 500

 WARNING: Total expenses exceed the budget.
```

Expense & Budget Summary:

```
> Please enter your choice: 3

     ==== Expense & Budget Summary ===

> Total Expenses: 900.0
> Budget Status: WARNING: Total expenses exceed the budget.
```

Exit Program:

```
> Please enter your choice: 4
 === Exiting System... ===
```